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Stat 201 Major Exam I Term 242

Original exam date: 15 February 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes Original course: STAT 319

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1. The given figure provides the stem-and-leaf diagram for 25 observations on batch yields from a chemical process. If stem: Tens digits. Leaf: Ones digits, then what is the maximum value of the data?

Stem	Leaf
6	1 3 4 5 5 6
7	0 1 1 3 5 7 8 8 9
8	1 3 4 4 7 8 8
9	2 3 5

(a) 0.95

(b) 95.0

(c) 90.5

(d) 9.50

(e) 905

2. For a discrete random variable X, the cumulative distribution function is given as:

$$F(x) = \begin{cases} 0, & x < -2 \\ 0.2, & -2 \le x < 0 \\ 0.7, & 0 \le x < 2 \\ 1, & x \ge 2 \end{cases}$$

What is the probability that x is exactly equal to 2?

(a) 1.0

(b) 0.2

(c) 0.3

(d) 0.5

(e) 0.1

3. Suppose that $P(A \mid B) = 0.4$ and $P(B) = 0.5$. Determine $P(A' \cap B)$.

(a) 0.9

(b) 0.2

(c) 0.1

(d) 0.3

(e) 0.5

4. If A, B, and C are mutually exclusive events with $P(A) = 0.2$, $P(B) = 0.3$, and $P(C) = 0.4$. Determine $P(A \cap B \cap C)$.

(a) 0.976

(b) 0.9

(c) 0.0

(d) 0.024

(e) 1.0

5. Contamination is a problem in the manufacture of optical storage disks (CDs). The number of particles of contamination that occur on an optical disk has a Poisson distribution, and the average number of particles per centimeter squared of media surface is 0.1. The area of a disk under study is 100 squared centimeters. Find the probability that 12 particles occur in the area of a disk under study.

(a) 0.905

(b) 0.095

(c) 0.792

(d) 0.001

(e) 0.208

6. The rise time of a reactor is measured in minutes (and fractions of minutes). Let the sample space be positive, real numbers. We define the events A and B as: $A = \{x \mid x < 72.5\}$ and $B = \{x \mid x > 52.5\}$. Describe the event $A \cup B$.

(a) $\{x \mid x \leq 52.5\}$

(b) $\{x \mid x < 0\}$

(c) $\{x \mid 52.5 < x < 72.5\}$

(d) $\{x \mid x > 0\}$

(e) $\{x \mid x \geq 72.5\}$

7. Two new product designs are to be compared on the basis of revenue potential. Marketing feels that the revenue from design A can be predicted quite accurately to be \$3 million. The revenue potential of design B is more difficult to assess. Marketing concludes that there is a probability of 0.3 that the revenue from design B will be \$7 million, but there is a 0.7 probability that the revenue will be only \$2 million. What is expected value of the revenue from design B (in millions of dollars)?

(a) 3.50

(b) 5.25

(c) 3.00

(d) 6.50

(e) 2.29

8. Suppose that the mean of a sample data is 4, and all the data values are same as the sample mean. What is the computed value of sample standard deviation?

(a) 1.0

(b) 16.0

(c) 4.0

(d) 2.0

(e) 0.0

9. The "cold ignition time" of an automobile engine is being investigated by a gasoline manufacturer. The following times (in seconds) were obtained for a test vehicle: $1.75, 1.92, 2.62, 2.35, 3.09, 3.15, 2.53, 1.91$. What is the value of upper quartile (Q3) of the data?

(a) 1.060

(b) 2.9725

(c) 0.750

(d) 1.9125

(e) 2.685

10. Orders for a computer are summarized by the optional features that are requested as follows:

Number of Operational Features	Proportion of Orders
No optional features	0.3
One optional feature	0.5
More than one optional feature	0.2

What is the probability that an order does not request more than one optional feature?

(a) 0.2

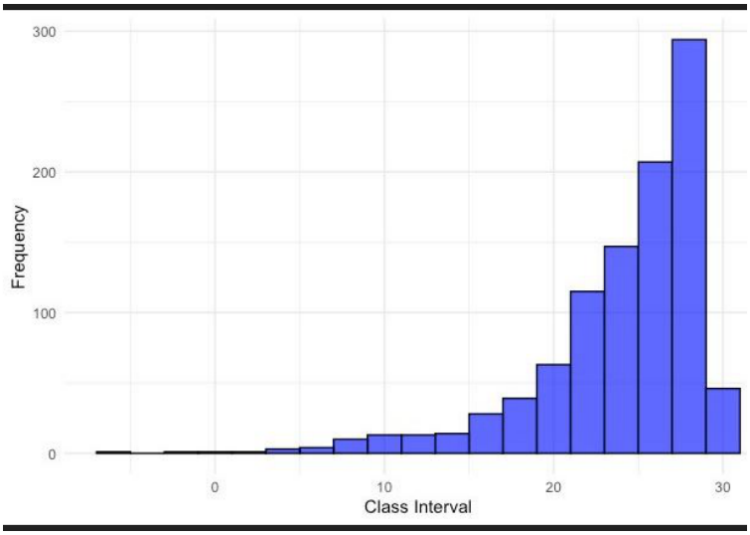
(b) 0.8

(c) 0.7

(d) 0.3

(e) 0.5

11. What statement is true for the histogram given here for a dataset?



(a) Mean $<$ Median

(b) Mean $=$ Median

(c) None

(d) Mean $= 2\times$ Median

(e) Mean $>$ Median

12. A manufacturing process has 100 customer orders to fill. Each order requires one component part that is purchased from a supplier. However, typically, 2% of the components are identified as defective, and the components can be assumed to be independent. If the manufacturer stocks 100 components, what is the probability that the 100 orders can be filled without reordering components?

(a) 0.0000

(b) 0.0885

(c) 0.98

(d) 0.13262

(e) 0.86738

13. Computer keyboard failures are due to faulty electrical connects (12%) or mechanical defects (88%). Mechanical defects are related to loose keys (27%) or improper assembly (73%). Electrical connect defects are caused by defective wires (35%), improper connections (13%), or poorly welded wires (52%). Find the probability that a failure is due to loose keys.

(a) 0.73

(b) 0.27

(c) 0.724

(d) 0.88

(e) 0.2376

14. Two web colors are used for a site advertisement. If a site visitor arrives from an affiliate, the probabilities of the blue or green colors are 0.8 and 0.2, respectively. If the site visitor arrives from a search site, the blue and green colors are 0.4 and 0.6, respectively. The proportions of visitors from affiliates and search sites are 0.3 and 0.7, respectively. What is the probability that a visitor is from a search site given that the blue ad was viewed?

(a) 0.28

(b) 0.24

(c) 0.52

(d) 0.50

(e) 0.54

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	Largest stem 9, largest leaf 5: maximum is 95.0.
2	**(c)**	$P(X = 2) = F(2) - F(2^-) = 1 - 0.7 = 0.3$.
3	**(d)**	$P(A \cap B) = 0.2$, so $P(A' \cap B) = 0.5 - 0.2 = 0.3$.
4	**(c)**	Mutually exclusive, so $A \cap B \cap C = \emptyset$ and the probability is 0.
5	**(b)**	$\lambda = 10$: $P(X = 12) = e^{-10}10^{12}/12! \approx 0.095$.
6	**(d)**	A and B overlap and together cover all positive reals: $\{x \mid x > 0\}$.
7	**(a)**	$E(B) = 7(0.3) + 2(0.7) = 3.5$.
8	**(e)**	All deviations from the mean are 0, so $s = 0$.
9	**(b)**	Sorted data, position $0.75(9) = 6.75$: $Q_3 = 2.62 + 0.75(0.47) = 2.9725$.
10	**(b)**	$P = 0.3 + 0.5 = 1 - 0.2 = 0.8$.
11	**(a)**	Left-skewed histogram (long left tail), so Mean $<$ Median.
12	**(d)**	$X \sim \text{Bin}(100, 0.02)$: $P(X = 0) = 0.98^{100} \approx 0.13262$.
13	**(e)**	$P = P(\text{mech})P(\text{loose} \mid \text{mech}) = 0.88(0.27) = 0.2376$.
14	**(e)**	$\dfrac{0.4(0.7)}{0.4(0.7) + 0.8(0.3)} = \dfrac{0.28}{0.52} \approx 0.54$.